AI/ML Engineer Roadmap

AI/ML Engineer Roadmap to Build Tools like ChatGPT, Gemini, Midjourney

Final Goal:

Create tools like ChatGPT (LLM), Gemini (Multimodal AI), Midjourney (Generative Image AI).

Phase 0: Mindset & Setup (Day 12)

- System Setup: VSCode, Python, Git, Anaconda
- Accounts: GitHub, Kaggle, HuggingFace, Google Colab

Phase 1: Programming Fundamentals (Days 320)

- Learn Python (Days 310): Variables, Loops, Functions, OOP, Libraries: NumPy, Pandas
- Projects: Calculator (CLI), Data Analysis (Titanic Dataset)

Phase 2: Math for ML (Days 2135)

- Topics: Linear Algebra, Probability, Calculus, Gradient Descent
- Projects: Linear Regression from Scratch, Matrix Calculator

Phase 3: Core Machine Learning (Days 3670)

- Algorithms: Linear/Logistic Regression, SVM, Trees, KNN, Clustering
- Evaluation: Accuracy, Confusion Matrix, ROC
- Libraries: scikit-learn, matplotlib, seaborn
- Projects: House Price Prediction, Spam Classifier

Phase 4: Deep Learning (Days 71110)

- Topics: Neural Nets, CNN, RNN, LSTM, Transformers
- Tools: TensorFlow or PyTorch
- Projects: MNIST Classifier, Sentiment Analysis (IMDb)

Phase 5: LLMs & NLP (Days 111150)

- Concepts: Tokenization, Embeddings, Attention, GPT/BERT
- Projects: Q&A Bot with BERT, GPT-2 Fine-tune

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Phase 6: Generative AI (Days 151180)

- Topics: GANs, Diffusion, CLIP
- Projects: Text-to-Image Generator, Style Transfer App

Phase 7: Real Tools (Days 181210)

- Projects: ChatGPT Clone (LangChain), Midjourney Clone, Voice Assistant

Phase 8: MLOps & Deployment (Days 211230)

- Tools: Docker, FastAPI, Gradio, Streamlit
- Projects: Chatbot Deployment, AI API

Phase 9: Portfolio (Days 231240)

- GitHub Profile, Docs, Portfolio Website

Tech Stack:

Python, scikit-learn, TensorFlow, PyTorch, HuggingFace, LangChain, Diffusers, CLIP, FastAPI, Gradio, Docker

Want daily roadmap? Say: Yes, make daily roadmap for 8 months.